Project Design Phase
Proposed Solution Template

Date	30 October 2026
Team ID	
Project Name	Automated Network Request Management
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	**Problem Statement (Problem to be solved)**	Organizations receive network service requests (new connections, existing connections and related services) from employees and other users. When these requests are handled without a structured process, the following problems occur: • incomplete request information and repeated manual data entry; • difficulty tracking request status and unclear responsibility for tasks; • delays and limited visibility caused by manual approval handling; • no timely acknowledgement or resolution notification; • inconsistent validation of the information users submit.
2.	**Idea / Solution description**	A ServiceNow scoped low-code application, built in App Engine Studio on a Personal Developer Instance, that organizes the whole request lifecycle: • **Request:** users submit a Network Request catalog item in the Service Portal; required fields prevent incomplete submissions, Existing ID appears only for Existing connections, and requester details can be auto-populated. • **Process:** a Flow Designer flow stores the request in a custom Database Tables table, emails an acknowledgement (request number, two business working days) and asks for approval; Approved or Rejected outcomes update the record. • **Track:** a separate Network Task Table holds the task and its Work Status. • **Resolve:** when Work Status changes to Completed, a second flow sends a 'Request has been Resolved' email. This is a working prototype of the lifecycle, not a production network-provisioning platform.
3.	**Novelty / Uniqueness**	The project is not a new technology; its value lies in how native platform features are combined for one focused process: • one platform covers the form, validation, workflow, approval, tasks and email, with no separate application server or database; • request approval state (Approval Status) and task progress (Work Status) are kept as separate concepts in separate tables; • approval records are shown on the request record through a related list; • two-stage email communication: acknowledgement when the request is created and notification when the work is completed.
4.	**Social Impact / Customer Satisfaction**	The intended users are employees who request network services and the teams who approve and fulfil them. Expected benefits are less repeated typing, fewer incomplete requests, a visible request number and status, and timely acknowledgement and resolution messages. No user survey or satisfaction measurement has been carried out, so satisfaction is not quantified; it should be assessed with real users in future work.
5.	**Business Model (Revenue Model)**	This is an academic prototype with no revenue model. Possible value for an organization is internal: less manual effort, faster handling and clearer tracking for its IT or network service desk. Possible future options (not implemented) include offering it as a reusable scoped application or as an implementation service for ServiceNow customers. Production use would require suitable ServiceNow licensing and a cost estimate, which have not been prepared.
6.	**Scalability of the Solution**	The application is organized as catalog, flows and custom tables on the ServiceNow platform, so new catalog items, request categories (for example VPN access or device provisioning), flows and tables can be added to the same scoped application. Planned enhancements include automatic task creation after approval, Integration Hub connections to network provisioning tools, dashboards, and update-set based deployment to other instances. Scalability has not been tested: no load or performance testing has been done, and capacity depends on the ServiceNow instance and platform limits.